

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	Haresh Kumar N L	Reg. No.:	192425009
Section / Batch:	AI&ML	Date:	24/07/2026

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1	50
		Q2	50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Bubble Sort – Large Element Drift Analysis

Apply the Brute Force technique using Bubble Sort on the dataset [13, 41, 19, 36, 8, 50]. Clearly interpret the problem and explain how Bubble Sort gradually moves larger elements toward their correct positions. Perform the sorting pass by pass and show all intermediate arrays. Count the total number of comparisons and swaps performed. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the algorithm's operational efficiency.

Code of Conduct

I Haresh Kumar N L (192425009) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Haresh

Bubble Sort:-

Definition:-

- Repeatedly compare adjacent elements and swap them if they are in the wrong order.
- After each Pass, the largest element "bubbles up" to the end of the array.
- Pass i ($0 \leq i \leq n-2$) can be represented as:

$$A_0 \dots A_i \hookrightarrow A_{i+1} \dots A_{n-i-1} \mid A_{n-i} \leq \dots \leq A_{n-1}$$

This means the last i elements are already sorted after each pass.

Algorithm:-

- Start at the beginning of the array.
- Compare each pair of adjacent elements from left to right.
- If the left element is greater than the right element, swap them.
- Continue until the end of the array.
- Repeat the process for the remaining unsorted part of the array until no swaps are needed.

Problem Given:-

[13, 41, 19, 36, 8, 50]

Initial Array :- [13, 41, 19, 36, 8, 50]

Phase 1:-

13, 41, 19, 36, 8, 50 ; 13 > 41, No swap increment by 1
i j

↓

13, 41, 19, 36, 8, 50 ; 41 > 19, swap 41 and 19
i j

↓

13, 19, 41, 36, 8, 50 ; 41 > 36, swap 41 and 36
i j

↓

13, 19, 36, 41, 8, 50 ; 41 > 8, swap 41 and 8
i j

↓

13, 19, 36, 8, 41, 50 ; 41 > 50, No swap
i j

Phase 2:-

13, 19, 36, 8, 41, 50 ; 13 > 19, No swap increment by 1
i j

↓

13, 19, 36, 8, 41, 50 ; 19 > 36, No swap increment by 1
i j

↓

13, 19, 36, 8, 41, 50 ; 36 > 8, swap 36 and 8
i j

↓

13, 19, 8, 36, 41, 50 ; 36 > 41, No swap increment by 1
i j

Phase 3:-

13, 19, 8, 36, 41, 50 ; 13 > 19, No swap increment by 1
i ; ↓

13, 19, 8, 36, 41, 50 ; 19 > 8, swap 19 and 8
i ; ↓

13, 8, 19, 36, 41, 50 ; 19 > 36, No swap
i ;

Phase 4:-

13, 8, 19, 36, 41, 50 ; 13 > 8, swap 13 and 8
i ; ↓

8, 13, 19, 36, 41, 50 ; 13 > 19, No swap
i ;

Phase 5:-

8, 13, 19, 36, 41, 50 ; 8 > 13, No swap
i ;

Total comparisons = 15

Total swap = 6

Original Array:-

[13, 41, 19, 36, 8, 50]

Sorted Array:-

[8, 13, 19, 36, 41, 50]

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)	4	4	4	4	4	40
Q2 (30 Marks)	4	4	4	4	4	30
Q3 (30 Marks)	4	3	3	2	2	22
Overall Total (100)						92

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____